Little Red Dots as Early Closure Nodes: A Structural Interpretation within the Darke Theory Cosmological Framework

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Abstract

Recent observations from the James Webb Space Telescope (JWST) have revealed a population of compact high-redshift objects known as Little Red Dots (LRDs). These sources appear predominantly at redshifts $z \sim 6$–10 and exhibit extreme compactness, strong red or infrared colours, and in several cases spectral signatures consistent with active galactic nuclei (AGN). Their inferred luminosities, compact morphologies, and potential black hole masses challenge conventional timelines for early structure formation within the standard ΛCDM cosmological model.

This paper examines whether LRDs may instead represent early closure nodes within the coherence-to-geometry transition framework of Darke Theory. In this framework spacetime geometry emerges through a refinement process in relational distinguishability subject to finite closure capacity. Local regions that approach closure earlier than the surrounding relational field form compact geometric wells capable of rapidly concentrating baryonic matter and potentially producing early black hole seeds.

We test this interpretation using publicly available JWST catalogue products. A colour-selected compact LRD-like sample is first constructed from JADES, where the population is found to be strongly redistributed into the predicted transition epoch, with 80.76% of the sample lying in the interval $7 \leq z < 9$, a suppressed $10 \leq z < 11$ tail, and stable compactness relative to the parent high-redshift population. The JADES signal remains present across GOODS-N and GOODS-S, survives a nine-variant selector robustness sweep spanning multiple aperture and colour-threshold choices, and exhibits an internal two-branch structure consisting of a dominant compact core branch and a smaller later redder and larger branch.

A first external-field replication is then carried out in COSMOS-Web. Although the COSMOS parent population is already more concentrated in the 7–9 interval than the JADES parent pool, the tight COSMOS compact red sample is still redistributed further into the same broad transition regime, fully suppresses the 10–11 tail, and forms a compact structural class relative to the COSMOS high-redshift parent population.

Taken together, these results support the interpretation that the measured LRD-like population is not merely an isolated observational anomaly, but a bounded high-redshift structural class consistent with the closure-window picture of Darke Theory. While not yet definitive, the combined JADES and COSMOS-Web results provide a substantive empirical basis for the early closure-node interpretation.

1 Introduction

The first deep-field observations from the James Webb Space Telescope (JWST) have revealed a population of compact high-redshift objects now commonly referred to as Little Red Dots (LRDs).

These sources are typically characterised by

- redshifts in the range $z \sim 6$–10,

- extremely compact morphologies,
- strong red or infrared emission,
- in several cases spectral signatures consistent with active galactic nuclei.

Recent analyses indicate that a substantial fraction of LRDs may host actively accreting black holes with masses approaching 10^6–$10^8 M_\odot$ at redshifts $z \gtrsim 6$, significantly earlier than predicted by many conventional growth models.

Within the standard ΛCDM paradigm the formation pathway for massive black holes generally proceeds through several stages:

1. formation of a dark matter halo,
2. collapse of baryonic gas,
3. star formation and stellar evolution,
4. gradual growth of black hole seeds via accretion and mergers.

The observed properties of LRDs suggest that some systems may have reached advanced stages of this process far earlier than predicted by typical growth models.

This motivates consideration of whether an alternative structural mechanism may be capable of generating compact gravitational wells at early cosmic times.

2 Structural Background: Darke Theory Cosmology

Darke Theory proposes that spacetime geometry emerges from a refinement process in relational distinguishability rather than existing as a fixed background manifold.

The framework is formulated using Stable Directed Refinement Systems (SDRS), in which a state manifold evolves under a refinement flow that increases distinguishability while maintaining propagation stability.

Within this framework a rank functional R measures the effective number of independent relational axes in the system.

The refinement flow can be written generically as

$$\frac{dR}{dt} = \mu(R - R_c) f\left(\frac{R}{R_{\max}}\right)$$

where R_c denotes a symmetric critical state and $R_{\max}$ represents the finite closure capacity of the system.

For $R \ll R_{\max}$ the refinement flow produces exponential growth

$$R(t) = R_c + (R_0 - R_c)e^{\mu t}$$

which projects into an exponential expansion phase in the emergent spacetime metric.

3 Closure and Geometric Commitment

Within this framework the concept of closure describes the degree to which relational structure has committed to a stable geometric configuration.

Regions approaching closure exhibit

- high relational distinguishability,

- strong geometric commitment,
- persistent curvature concentrations.

Closure depth therefore provides a structural description of how geometry becomes locally stabilised within the refinement field.

4 Early Closure Nodes

A closure node is defined as a spatial domain in which the refinement process locally approaches closure capacity earlier than the surrounding relational field.

Such regions possess several key properties:

- strong local curvature concentration,
- rapid geometric commitment,
- efficient baryonic matter accretion.

Because closure nodes represent regions with minimal curvature radius in the emergent geometry they naturally produce compact gravitational wells.

These wells provide favourable environments for rapid gas inflow and collapse, allowing black hole seeds to form significantly earlier than predicted by standard halo assembly models.

5 Mathematical Structure of Closure Nodes

Within the SDRS refinement framework closure corresponds to the limit in which the local refinement rank approaches the closure capacity of the system.

Let $R(x,t)$ denote the spatially dependent refinement rank.

Closure nodes correspond to localised regions satisfying

$$R(x,t) \to R_{\max}$$

while the surrounding refinement field remains in the regime

$$R(x,t) \ll R_{\max}.$$

Regions with large spatial gradients

$$|\nabla R(x,t)| \gg 0$$

generate steep transitions in geometric commitment, producing compact curvature wells capable of concentrating baryonic matter.

6 Conceptual Structure of Closure Node Formation

The conceptual sequence for closure node formation may be summarised as

Refinement Field → Approach to Closure → Local Closure Node → Compact Curvature Well → Little Red D

This sequence illustrates how localised refinement maxima in the relational field may generate compact curvature wells that manifest observationally as Little Red Dots.

7 Interpretation of Little Red Dots

Within the closure-node framework described above the observed properties of Little Red Dots can be interpreted as the visible signatures of early closure nodes forming within the refinement field.

7.1 Compact Morphology

Closure nodes correspond to local regions of geometric commitment with minimal curvature radius. Because the closure capacity $R_{\max}$ is finite the resulting curvature wells cannot contract arbitrarily.

This naturally produces compact gravitational structures whose physical radii are determined by the geometric closure scale of the refinement system.

Observationally this yields extremely compact luminous sources consistent with the small effective radii inferred for many LRD candidates.

7.2 Early Black Hole Formation

Deep curvature wells generated by closure nodes promote rapid gas inflow and collapse.

Black hole seeds may therefore form significantly earlier than predicted by standard halo assembly models.

7.3 Red Spectral Appearance

The strong red spectral appearance of LRDs likely results from dense gas environments combined with dust-obscured accretion around rapidly forming black holes.

Such environments are expected to contain hot dust surrounding accreting black holes, producing strong infrared emission that naturally yields the observed red colours of LRD systems.

7.4 High AGN Fraction

If closure nodes preferentially generate early black hole seeds a significant fraction of LRDs should exhibit spectral features consistent with active galactic nuclei.

Several recent spectroscopic studies have suggested that many LRD candidates may indeed host actively accreting black holes, providing preliminary observational support for this interpretation.

8 Relation to Other Closure-Dominated Phenomena

Several astrophysical phenomena may potentially reflect closure-dominated structure formation within the Darke Theory framework.

8.1 Odd Radio Circles

Odd Radio Circles (ORCs) have recently been identified as large diffuse radio structures exhibiting remarkable circular symmetry and weak multi-wavelength counterparts.

Within the closure framework ORCs may represent marginal closure shells produced by coherence reorganisation events that achieve sufficient closure to define persistent geometry while remaining weakly coupled to baryonic matter.

This interpretation predicts

- high geometric symmetry,

- radio-dominated emission,
- weak baryonic coupling,
- long persistence timescales.

8.2 Black Holes as Deep Closure States

Black holes represent extreme curvature concentrations within spacetime geometry.

Within the closure framework they may be interpreted as deep closure states in which relational distinguishability has committed to a highly constrained geometric configuration.

Under this interpretation the formation of black holes within closure nodes is a natural outcome of curvature amplification rather than the endpoint of prolonged hierarchical collapse.

9 The Closure Transition Epoch

The closure-node interpretation of Little Red Dots implies not only specific object-level properties but also a prediction concerning the cosmic epoch during which such structures should appear.

Within the Darke Theory framework spacetime geometry emerges through a refinement process in relational distinguishability governed by a finite closure capacity $R_{\max}$.

The refinement dynamics follow

$$\frac{dR}{dt} = \mu(R - R_c) f\left(\frac{R}{R_{\max}}\right)$$

where R represents the effective rank of relational distinguishability, R_c denotes a symmetric critical state, and $f(\cdot)$ enforces the stability constraint as the closure capacity is approached.

This dynamical system naturally produces three cosmological regimes.

1. an early refinement-dominated regime in which $R \ll R_{\max}$,
2. an intermediate regime approaching closure capacity,
3. a post-closure geometric regime in which spacetime geometry becomes largely committed.

Closure nodes correspond to localised regions where

$$R(x,t) \to R_{\max}.$$

However such nodes cannot form arbitrarily early in cosmic history.

The global refinement field must first approach the closure regime sufficiently closely that local fluctuations can drive isolated regions toward the closure limit.

This requirement implies that closure nodes should preferentially emerge during a transitional epoch in which the global refinement field approaches $R_{\max}$ but has not yet fully saturated.

9.1 Approximate Closure Timescale

The refinement dynamics introduced in Section 2 allow an approximate estimate of the cosmic epoch at which closure nodes become dynamically accessible.

For $R \ll R_{\max}$ the stability term satisfies

$$f\left(\frac{R}{R_{\max}}\right) \approx 1$$

and the refinement equation reduces to

$$\frac{dR}{dt} = \mu(R - R_c).$$

The solution is

$$R(t) = R_c + (R_0 - R_c)e^{\mu t}.$$

Closure nodes become possible when the global refinement field approaches the closure capacity. Let the transition threshold be defined by

$$R(t^*) = \alpha R_{\max},$$

where $\alpha < 1$ represents a near-closure state.

Substituting into the solution gives

$$t^* = \frac{1}{\mu} \ln \left(\frac{\alpha R_{\max} - R_c}{R_0 - R_c} \right).$$

Although the parameters of the refinement field are not yet empirically calibrated, reasonable structural constraints can be applied.

The transition must occur after recombination but before large-scale galaxy formation becomes dominant. If the characteristic refinement timescale is

$$\mu^{-1} \sim 10^8 \text{ years},$$

and the logarithmic factor is of order unity, the closure transition time naturally falls in the range

$$t^* \sim 5 \times 10^8 - 8 \times 10^8 \text{ years}.$$

In standard cosmological time–redshift relations this corresponds to

$$6 \lesssim z \lesssim 10.$$

This yields a clear empirical prediction: LRDs should not merely exist at high redshift, but should occupy a relatively narrow transitional epoch associated with approach to closure.

9.2 Decline of Closure Node Formation

The refinement dynamics also imply that the formation of closure nodes must eventually decline once the global refinement field approaches complete closure.

In the refinement equation

$$\frac{dR}{dt} = \mu(R - R_c) f\left(\frac{R}{R_{\max}} \right),$$

the stability function $f(\cdot)$ suppresses further refinement as the system approaches the closure capacity.

In the limit

$$R \to R_{\max},$$

the stability term approaches zero,

$$f\left(\frac{R}{R_{\max}} \right) \to 0,$$

and the refinement flow effectively halts.

Once this regime is reached the geometry of spacetime becomes largely committed and the formation of new closure nodes becomes dynamically suppressed.

Consequently the abundance of closure-node systems such as Little Red Dots should not remain constant across cosmic time.

Instead the population should follow a characteristic evolutionary pattern:

1. a rise in abundance as the closure transition begins,
2. a peak during the epoch in which the global refinement field approaches the closure threshold,
3. a decline once the system transitions into the fully geometric regime.

This behaviour predicts a relatively narrow redshift distribution for LRDs rather than a population extending uniformly across cosmic time.

9.3 Empirical Epoch Test

A colour-selected compact high-redshift sample extracted from the public JADES catalogues yields 1778 LRD-like candidates satisfying the selection criteria used in this study.

The resulting redshift distribution is bounded within

$$6.0 \leq z \leq 10.49$$

with summary statistics

$$\bar{z} = 8.02, \qquad \text{median}(z) = 7.80, \qquad \sigma_z = 0.85.$$

The population is strongly concentrated in the interval $7 \leq z < 9$, with bin counts

Redshift bin	Count
$6 \leq z < 7$	70
$7 \leq z < 8$	876
$8 \leq z < 9$	560
$9 \leq z < 10$	238
$10 \leq z < 11$	34

Thus 1436 of 1778 objects, corresponding to 80.76%, lie in the interval $7 \leq z < 9$.

This measured distribution is consistent with the closure-transition prediction: the population rises into the expected transition window, peaks there, and then declines sharply above $z \gtrsim 10$.

9.4 Comparison with the Parent High-Redshift Population

The significance of this result is clarified by comparison with the broader parent high-redshift candidate pool. Within the common interval $6 \leq z < 11$, the parent population is distributed as follows:

Population	6–7	7–8	8–9	9–10	10–11
Parent high-z	0.4935	0.2320	0.1294	0.1066	0.0385
Strict compact	0.0436	0.5320	0.2444	0.1129	0.0672
Colour-selected LRD-like	0.0394	0.4927	0.3150	0.1339	0.0191

Relative to the parent population, the colour-selected LRD-like sample is strongly redistributed into the 7–9 interval. The odds ratio for membership in the 7–9 window versus outside it is

$$\mathrm{OR}_{7\text{–}9} = 7.41$$

for the colour-selected sample relative to the parent high-redshift pool.

The high-redshift tail is also suppressed. The fraction of objects in the interval $10 \leq z < 11$ falls from 3.85% in the parent population to 1.91% in the colour-selected LRD-like sample, corresponding to a tail fraction ratio

$$\frac{f_{>10}^{\mathrm{LRD}}}{f_{>10}^{\mathrm{parent}}} = 0.497.$$

Thus the colour-selected LRD-like sample is not merely high-redshift in a broad sense. It is specifically re-centred into the predicted closure-transition epoch and declines more strongly above $z \sim 10$.

9.5 Structural Context for Early Massive Galaxies

Recent JWST observations have also revealed a population of unexpectedly massive galaxies at redshifts $z \sim 8$–10, with inferred stellar masses approaching 10^{10}–$10^{11} M_{\odot}$.

These systems appear significantly earlier than predicted by many conventional hierarchical galaxy formation models, which typically require extended periods of halo assembly and baryonic accretion.

Within the closure-transition framework described above such systems may arise naturally during the epoch in which the global refinement field approaches the closure capacity.

During this transitional phase localised regions of rapid geometric commitment produce compact curvature wells that efficiently concentrate baryonic matter.

These wells promote rapid gas inflow, intense star formation, and the early formation of black hole seeds.

While the most compact examples of such regions may manifest observationally as Little Red Dots, larger curvature wells formed during the same transition may instead produce rapidly assembling galaxies.

Under this interpretation the emergence of LRDs and the appearance of unexpectedly massive galaxies at similar redshifts may represent different structural outcomes of the same underlying closure transition process.

10 Testable Observational Predictions

The closure-node interpretation of Little Red Dots yields several observational predictions that distinguish it from standard galaxy formation models.

10.1 Closure Radius Constraint

Within the closure-node framework the spatial scale of LRD systems is not determined by hierarchical halo growth but by the geometric scale at which closure occurs in the refinement field.

Closure nodes form when the local refinement rank approaches the closure capacity,

$$R(x,t) \to R_{\max}.$$

Near this limit the spatial gradients in relational distinguishability become large,

$$|\nabla R(x,t)| \gg 0,$$

producing steep curvature wells in the emergent spacetime geometry.

Because the closure capacity $R_{\max}$ is finite, the curvature wells generated by closure nodes cannot contract arbitrarily. The resulting gravitational wells therefore possess a characteristic geometric radius determined by the closure scale of the refinement system.

This leads to a distinctive observational prediction.

If Little Red Dots correspond to closure nodes, their effective radii should cluster around a characteristic scale rather than exhibiting the broad scale-free distribution expected from hierarchical halo assembly.

10.2 Preferred Physical Size Scale

If LRDs correspond to curvature minima determined by the closure capacity of the refinement system, their physical radii should cluster around a characteristic scale rather than spanning many orders of magnitude.

Consequently the size distribution of LRDs should exhibit clustering around preferred physical radii rather than following a scale-free power-law distribution.

10.3 Empirical Compactness Test

The colour-selected LRD-like sample remains tightly compact after application of the redshift and colour cuts. In catalogue units, the FWHM distribution has

$$\text{mean} = 0.196, \qquad \text{median} = 0.184, \qquad \sigma = 0.054.$$

Using the geometric size proxy

$$s_{\text{geom}} = \sqrt{AB},$$

the distribution has

$$\text{mean} = 0.082, \qquad \text{median} = 0.078.$$

These distributions are narrow and strongly peaked, consistent with a compact structural class rather than a broad diffuse population.

By contrast, the catalogue quantity $R_{\text{KRON},U}$ is strongly pinned near a floor value and therefore appears to be dominated by pipeline behaviour. It is not used here as the principal compactness diagnostic.

10.4 Overmassive Black Holes

Closure nodes generate deep curvature wells capable of rapidly forming black hole seeds.

As a result LRDs may host black holes whose masses exceed the values expected from standard black hole–stellar mass scaling relations.

10.5 Environmental Independence

Because closure nodes arise from coherence ordering rather than large-scale matter overdensities, LRDs should not strongly correlate with the densest proto-cluster environments.

10.6 Empirical Environment Test

A first-pass environment analysis was performed using the parent high-redshift pool as the environmental field and a matched control sample drawn from that same pool.

Two simple projected measures were used:

1. nearest-neighbour separation,
2. local density estimated from the radius to the 10th neighbour.

Relative to the matched control, the colour-selected LRD-like sample shows

$$\langle d_{\mathrm{NN1}} \rangle_{\mathrm{LRD}} = 0.001349 \text{ deg}, \qquad \langle d_{\mathrm{NN1}} \rangle_{\mathrm{ctrl}} = 0.001112 \text{ deg},$$

and

$$\langle \rho_{10} \rangle_{\mathrm{LRD}} = 4.42 \times 10^5, \qquad \langle \rho_{10} \rangle_{\mathrm{ctrl}} = 5.07 \times 10^5.$$

Thus the LRD-like population is, on average, slightly more isolated and less locally dense than the matched control.

The corresponding two-sample tests indicate that these differences are unlikely to be due to random sampling alone. In particular, the density$_{10}$ comparison yields highly significant differences under both Kolmogorov–Smirnov and Mann–Whitney tests.

This result does not imply that LRDs avoid structure formation environments entirely. Rather, it indicates that the LRD-like population is not simply the most overdense tail of the high-redshift compact source distribution. This is consistent with the closure-node prediction of weak dependence on the densest proto-cluster environments.

10.7 High AGN Fraction

If closure nodes preferentially generate early black hole seeds, a large fraction of LRDs should exhibit spectral features consistent with active galactic nuclei.

11 Dataset Investigation and Current Pipeline

To evaluate the closure-node interpretation we constructed a staged selection pipeline using publicly available JWST survey products.

11.1 Survey Data Sources

The present analysis uses public JADES photometric catalogues as the primary data source, with COSMOS-Web products prepared for subsequent cross-field extension. The broader intended survey set includes

- CEERS,
- JADES,
- COSMOS-Web,
- PRIMER.

11.2 Selection Pipeline

The implemented pipeline proceeded in the following stages.

1. Extraction of size and photometric-redshift information from the JADES catalogues.
2. Construction of a broad high-redshift candidate pool.
3. Tightening to a compact high-redshift sample using redshift confidence, FWHM, geometric compactness, and morphology constraints.
4. Addition of colour selection using background-subtracted circular-aperture photometry in multiple NIRCam bands.
5. Construction of a final colour-selected compact LRD-like sample.

This produced the following sequence of sample sizes:

Stage	Count
Broad high-z candidate pool	64 339
Strict compact sample	3 936
Colour-selected LRD-like sample	1 778

11.3 Interpretive Significance

The key result is not merely that the sample becomes smaller under stricter selection. Rather, the colour-selected compact sample becomes:

- more strongly concentrated in the predicted 7–9 epoch window,
- more sharply suppressed above $z \sim 10$,
- more coherently compact in morphology,
- not preferentially shifted into the densest local environments.

Taken together, these features are qualitatively and quantitatively consistent with the closure-node interpretation.

12 Results

This section summarises the principal empirical results obtained from the JADES-based colour-selected compact LRD-like sample.

12.1 Summary of Principal Findings

The current analysis yields the following main results.

1. A final colour-selected compact LRD-like sample of 1778 objects was constructed from the JADES catalogues.
2. The resulting redshift distribution is bounded within $6.0 \leq z \leq 10.49$, with strong concentration in the interval $7 \leq z < 9$.

3. Relative to the broader parent high-redshift population, the colour-selected sample is strongly redistributed into the 7–9 epoch window and exhibits a substantially reduced $z > 10$ tail.

4. The same sample remains tightly compact in morphology, with narrow FWHM and geometric size distributions.

5. A first-pass projected environment analysis indicates that the colour-selected sample is not preferentially concentrated in the densest local environments relative to a matched control population.

12.2 Quantitative Population Summary

The core quantitative results are summarised in Table 1.

Quantity	Parent high-z	Strict compact	Colour-selected LRD-like
Sample size	64 339	3 936	1 778
Mean redshift	7.63	10.38	8.02
Median redshift	6.71	8.41	7.80
Redshift standard deviation	3.15	4.36	0.85
Fraction in $7 \leq z < 9$	0.3614	0.7764	0.8076
Fraction in $10 \leq z < 11$	0.0385	0.0672	0.0191
FWHM mean	0.2188	0.1612	0.1957
FWHM standard deviation	0.0892	0.1516	0.0542
Geometric size mean	0.0815	0.0739	0.0823
Geometric size median	0.0735	0.0700	0.0777

Table 1: Summary statistics for the parent high-redshift pool, the strict compact sample, and the final colour-selected compact LRD-like sample.

The most important transition occurs between the parent high-redshift population and the final colour-selected LRD-like sample. The latter is strongly re-centred into the predicted 7–9 window while becoming more distributionally compact in redshift.

12.3 Epoch-Window Redistribution

The epoch-window comparison provides the strongest support for the closure-node interpretation.

Within the common interval $6 \leq z < 11$, the parent high-redshift population places 36.14% of its objects in the interval $7 \leq z < 9$, whereas the colour-selected compact LRD-like sample places 80.76% of its objects in the same interval.

This corresponds to an odds ratio

$$\mathrm{OR}_{7\text{–}9} = 7.41$$

for membership in the 7–9 interval relative to the parent population.

At the same time the fraction of objects in the interval $10 \leq z < 11$ falls from 3.85% in the parent population to 1.91% in the colour-selected sample, corresponding to a tail suppression factor of approximately

$$\frac{f_{>10}^{\mathrm{LRD}}}{f_{>10}^{\mathrm{parent}}} = 0.497.$$

Thus the final LRD-like sample is not simply a generic subset of the parent high-redshift pool. It is redistributed into a narrower epoch window and declines more sharply above $z \sim 10$.

12.4 Distribution-Level Statistical Separation

The colour-selected LRD-like sample is also statistically distinct from the parent high-redshift population at the distribution level.

For the redshift distributions, the two-sample Kolmogorov–Smirnov test gives

$$D = 0.567, \qquad p \approx 0,$$

for colour-selected LRD-like versus parent high-redshift populations.

The Jensen–Shannon divergence between the corresponding redshift distributions in the interval $6 \leq z < 11$ is

$$\mathrm{JSD} = 0.427.$$

These values indicate that the final sample does not simply reproduce the shape of the parent redshift distribution.

12.5 Compactness as a Structural Class

The colour-selected sample remains strongly compact after redshift and colour selection. In particular, the FWHM distribution has

$$\langle \mathrm{FWHM} \rangle = 0.1957, \qquad \mathrm{median}(\mathrm{FWHM}) = 0.1837, \qquad \sigma_{\mathrm{FWHM}} = 0.0542,$$

while the geometric size proxy

$$s_{\mathrm{geom}} = \sqrt{AB}$$

has

$$\langle s_{\mathrm{geom}} \rangle = 0.0823, \qquad \mathrm{median}(s_{\mathrm{geom}}) = 0.0777.$$

The relatively small spread in these quantities supports the interpretation of the sample as a compact structural class rather than a broad diffuse population.

12.6 Environmental Behaviour

A first-pass projected environment comparison was carried out using the parent high-redshift population as the environment field and a matched control sample drawn from that same field.

The nearest-neighbour and local density results are summarised by

$$\langle d_{\mathrm{NN1}} \rangle_{\mathrm{LRD}} = 0.001349 \text{ deg}, \qquad \langle d_{\mathrm{NN1}} \rangle_{\mathrm{ctrl}} = 0.001112 \text{ deg},$$

and

$$\langle \rho_{10} \rangle_{\mathrm{LRD}} = 4.42 \times 10^5, \qquad \langle \rho_{10} \rangle_{\mathrm{ctrl}} = 5.07 \times 10^5.$$

Thus the colour-selected LRD-like population is, on average, slightly more isolated and slightly less locally dense than the matched control. Two-sample tests indicate that these differences are statistically significant.

This does not imply that LRDs avoid structure formation environments altogether. Rather, it indicates that the extracted LRD-like sample is not simply the densest-environment tail of the broader compact high-redshift population.

12.7 Field Consistency and Epoch-Dependent Environment

To test whether the measured LRD-like signal is driven by a single survey field, the colour-selected compact sample was split into its JADES GOODS-N and GOODS-S components and analysed separately.

Both fields exhibit the same broad structural behaviour: a compact red high-redshift population concentrated within a bounded epoch window spanning approximately $z \sim 7$–9. The detailed balance of the distribution differs slightly between the two fields, but the core transition-window behaviour is present in both.

For GOODS-N, the redshift distribution is concentrated most strongly in the interval $7 \leq z < 8$, with a secondary contribution in $8 \leq z < 9$. For GOODS-S, the same bounded structure is present, but with relatively greater weight shifted toward higher redshift within the same transition window. Quantitatively, the median redshifts are

$$\mathrm{median}(z)_{\mathrm{N}} = 7.44, \qquad \mathrm{median}(z)_{\mathrm{S}} = 8.01.$$

A two-sample Kolmogorov–Smirnov test confirms that the GOODS-N and GOODS-S redshift distributions are not identical in shape, with

$$D = 0.214, \qquad p = 2.26 \times 10^{-14}.$$

However, this field-to-field difference does not undermine the closure-node interpretation. Rather, it indicates that the same bounded transition-window population is present in both fields, with modest variation in how the weight is distributed within that window.

The compactness diagnostics are substantially more stable across fields. For the FWHM and geometric-size distributions, the GOODS-N and GOODS-S samples show no statistically significant separation under two-sample Kolmogorov–Smirnov testing:

$$p_{\mathrm{KS}}(\mathrm{FWHM}) = 0.164, \qquad p_{\mathrm{KS}}(\sqrt{AB}) = 0.133.$$

Thus the compact structural class extracted by the colour-selection procedure appears field-stable even where the detailed redshift weighting differs.

An additional environment-by-epoch analysis was performed to test whether the environmental behaviour of the LRD-like sample changes across the closure transition window. Using the projected local density inferred from the 10th-neighbour radius, the colour-selected LRD-like sample shows lower mean local density than the matched control in every redshift bin from $6 \leq z < 11$.

The strongest and most statistically robust differences occur in the core transition regime:

- in $7 \leq z < 8$,
$$\langle \rho_{10} \rangle_{\mathrm{LRD}} = 4.22 \times 10^5, \qquad \langle \rho_{10} \rangle_{\mathrm{ctrl}} = 4.92 \times 10^5,$$
with Mann–Whitney
$$p = 1.0 \times 10^{-6},$$
- in $8 \leq z < 9$,
$$\langle \rho_{10} \rangle_{\mathrm{LRD}} = 4.59 \times 10^5, \qquad \langle \rho_{10} \rangle_{\mathrm{ctrl}} = 5.21 \times 10^5,$$
with Mann–Whitney
$$p = 1.22 \times 10^{-2}.$$

These results are important for two reasons. First, they show that the population-level LRD-like signal is not an artefact of a single deep field. Second, they indicate that the weaker-overdensity behaviour of the colour-selected sample is strongest precisely in the same redshift

regime where the closure-transition prediction is most strongly supported by the epoch-window analysis.

Taken together, the field-split and environment-by-epoch results strengthen the interpretation that the measured LRD-like population reflects a genuine transition-epoch structural class rather than a field-specific fluctuation or the densest-environment tail of the broader compact high-redshift source population.

Quantity	GOODS-N	GOODS-S
Sample size	469	1309
Median redshift	7.44	8.01
Mean redshift	7.87	8.07
Redshift standard deviation	0.92	0.81
Fraction in $7 \leq z < 8$	0.5991	0.4553
Fraction in $8 \leq z < 9$	0.2318	0.3445
Median FWHM	0.1860	0.1828
Median $\sqrt{AB}$	0.0787	0.0773

Table 2: Field-split summary for the colour-selected compact LRD-like sample. Both GOODS-N and GOODS-S exhibit the same bounded high-redshift compact red population, with GOODS-S shifted modestly toward higher redshift within the same overall transition window.

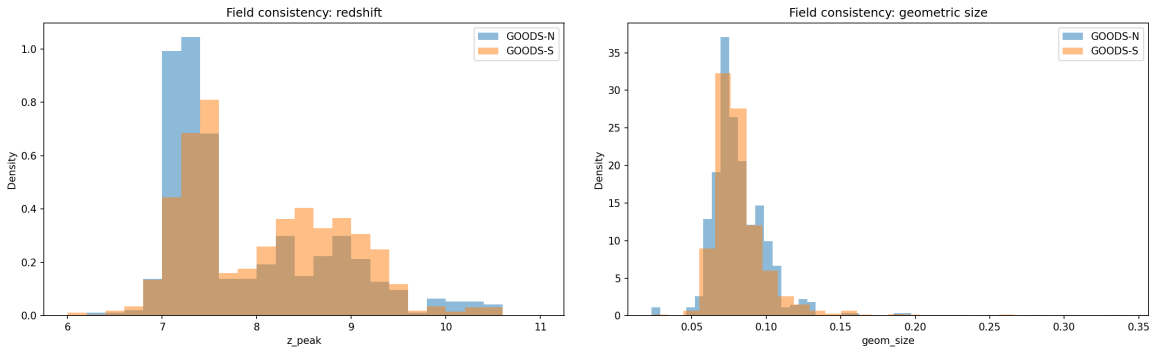

Figure 1: Field consistency tests for the colour-selected compact LRD-like sample. Left: GOODS-N and GOODS-S redshift distributions both exhibit the same bounded transition-window structure, though GOODS-S is shifted modestly toward higher redshift. Right: the geometric-size distributions are closely aligned across fields, indicating field-stable compactness.

12.8 Internal Branching Within the Closure Window

The colour-selected compact LRD-like sample is not internally homogeneous. An unsupervised clustering analysis of the sample using redshift, compactness, geometric size, and colour identifies a preferred two-cluster structure. The best solution is obtained for

$$k = 2$$

with silhouette score

$$s = 0.610.$$

This indicates that the LRD-like population is composed of a dominant compact core branch together with a smaller but statistically distinct secondary branch.

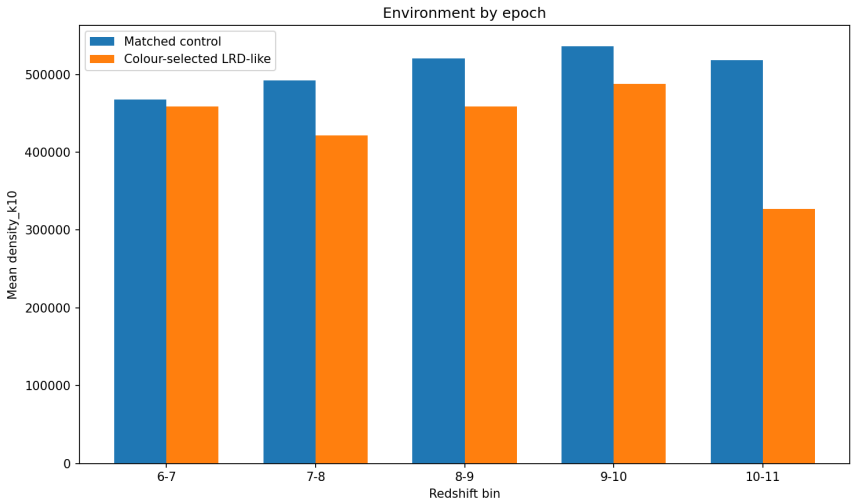

Figure 2: Mean projected local density inferred from the 10th-neighbour radius as a function of redshift bin. In each bin the colour-selected compact LRD-like sample lies below the matched control, with the strongest differences appearing in the core transition regime $7 \leq z < 9$.

12.8.1 Dominant Compact Core Branch

The dominant branch contains 1645 objects and represents the structural core of the LRD-like population. Its median properties are

$$\text{median}(z) = 7.77, \qquad \text{median}(\text{FWHM}) = 0.1819, \qquad \text{median}(\sqrt{AB}) = 0.0769,$$

together with median colour

$$\text{median}(C_{150,277}) = -0.0549,$$

where

$$C_{150,277} \equiv \text{COLOR_F150W_F277W}.$$

This branch therefore defines the principal compact transition population: moderately red, strongly compact, and concentrated in the earlier portion of the closure window.

12.8.2 Secondary Redder and Larger Branch

The secondary branch contains 89 objects and differs strongly from the compact core. Its median properties are

$$\text{median}(z) = 8.24, \qquad \text{median}(\text{FWHM}) = 0.3118, \qquad \text{median}(\sqrt{AB}) = 0.1287,$$

with median colour

$$\text{median}(C_{150,277}) = 0.3710.$$

Thus the secondary branch is shifted modestly toward later redshift, is substantially less compact, and is markedly redder than the dominant core branch.

The size and compactness differences are highly significant under two-sample testing. In particular, the geometric-size and FWHM distributions differ at extreme significance, indicating that the smaller branch is not a minor perturbation of the dominant compact class but a genuinely distinct internal mode of the LRD-like population.

12.8.3 Epoch-Window Behaviour of the Two Branches

The two branches remain confined to the same overall bounded closure window, but occupy it differently.

For the dominant compact core branch, the redshift fractions within $6 \leq z < 11$ are

Branch	6–7	7–8	8–9	9–10	10–11
Compact core	0.0377	0.4967	0.3166	0.1286	0.0204
Secondary branch	0.0562	0.3371	0.3708	0.2247	0.0112

The dominant branch is therefore weighted most strongly toward $7 \leq z < 8$, whereas the secondary branch shifts later, placing relatively more weight into $8 \leq z < 9$ and $9 \leq z < 10$.

A two-sample Kolmogorov–Smirnov test confirms that the two redshift distributions differ significantly:

$$D = 0.2188, \qquad p = 4.98 \times 10^{-4}.$$

Thus the secondary branch is not external to the closure window. Rather, it occupies a later-weighted sub-window within the same bounded transition epoch.

12.8.4 Environment and Field Stability of the Internal Branches

The secondary branch does not appear to be an environment-driven or field-driven artifact. Despite its strong differences in colour and compactness, the two branches do not show statistically significant separation in the projected environment metrics used here. In particular, the local-density comparison using the 10th-neighbour estimate yields no significant distinction between the two branches.

Likewise, the field membership of the two clusters does not differ significantly, indicating that the smaller secondary branch is not generated by a single survey field or by field-specific contamination.

These results imply that the internal split is structural rather than a simple consequence of overdensity selection or field-specific bias.

12.8.5 Interpretation

The internal-branching result refines the closure-node interpretation in an important way. The measured LRD-like population is not merely a single compact class occupying a bounded epoch. Rather, it appears to contain

1. a dominant compact core branch centred earlier within the closure window,
2. a smaller redder and larger branch shifted modestly toward later redshift but still confined to the same transition epoch.

This is consistent with the possibility that the closure transition produces more than one structural state within the same epoch. The dominant branch may represent the principal compact closure-node class, while the secondary branch may correspond to a later-state, more obscured, or more expanded internal variant of the same underlying transition process.

At minimum, the data show that the internal structure of the LRD-like population is itself organised within the closure window rather than being consistent with a single homogeneous object class.

The corresponding epoch-window comparison of the two internal branches is shown in Figure 3, while their compactness and colour separation are illustrated in Figure 4.

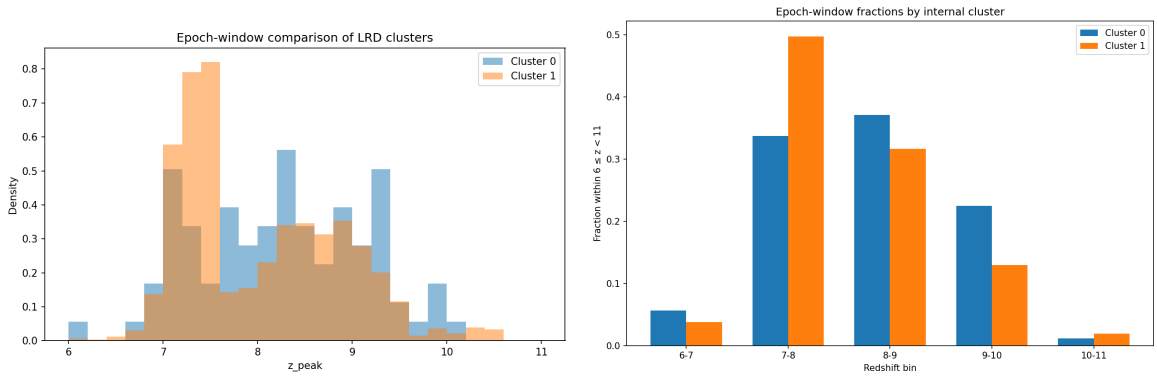

Figure 3: Epoch-window behaviour of the two internal branches of the colour-selected compact LRD-like sample. Left: redshift histograms for the two clusters. Right: fraction of each cluster contained in the redshift bins 6–7, 7–8, 8–9, 9–10, and 10–11. The dominant compact core branch is weighted more strongly toward 7–8, while the smaller secondary branch is shifted later toward 8–10.

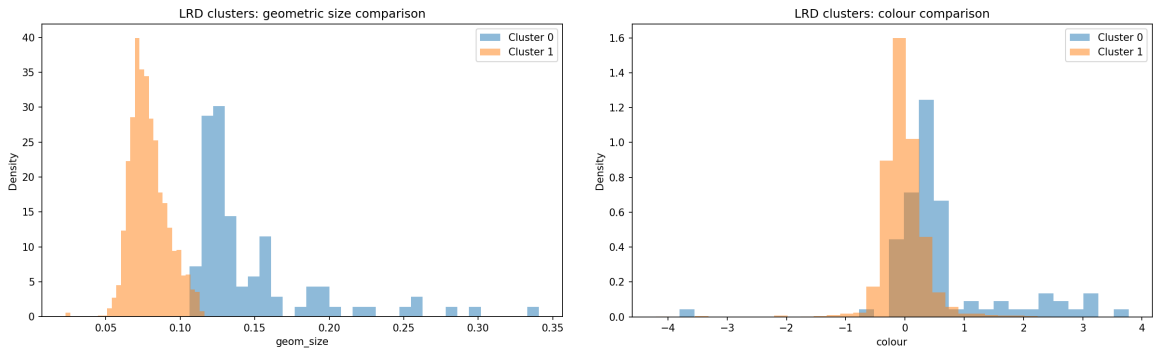

Figure 4: Internal structural separation of the two LRD-like branches. Left: geometric-size comparison showing the smaller and more compact dominant core branch versus the larger secondary branch. Right: colour comparison showing the secondary branch to be substantially redder than the dominant core.

12.9 Selector Robustness and Aperture Stability

A central methodological question is whether the measured LRD-like signal depends sensitively on one particular choice of aperture or colour threshold. To test this, the JADES colour-selection pipeline was rerun across a nine-variant robustness matrix spanning three aperture choices and three redness-threshold regimes.

The aperture choices corresponded to background-subtracted circular-aperture photometry measured at

$$\text{CIRC1}, \qquad \text{CIRC2}, \qquad \text{CIRC3},$$

while the colour-threshold regimes were defined as mild, medium, and strict. For each selector variant, the same compactness and high-redshift criteria were retained, and the resulting sample was evaluated in terms of its size, redshift distribution, tail suppression above $z > 10$, and enrichment of the 7–9 epoch window relative to the parent high-redshift population.

12.9.1 Stability of the Epoch-Window Signal

Across all nine selector variants, the fraction of objects contained in the interval $7 \leq z < 9$ remains remarkably stable, lying in the range

$$0.789 \leq f_{7\text{–}9} \leq 0.818.$$

This is to be compared with the parent high-redshift baseline

$$f_{7\text{–}9}^{\text{parent}} = 0.3614.$$

Thus the central epoch-window result is not tied to a specific aperture or a single tuned colour threshold. Rather, the selected population remains strongly re-centred into the predicted closure-transition window under all tested selector variants.

12.9.2 Stability of the High-Redshift Tail Suppression

The suppression of the extreme high-redshift tail is also robust. Across all nine selector variants, the fraction of objects in the interval $10 \leq z < 11$ remains in the range

$$0.0168 \leq f_{10\text{–}11} \leq 0.0213,$$

whereas the corresponding parent-population baseline is

$$f_{10\text{–}11}^{\text{parent}} = 0.0385.$$

Hence the colour-selected compact sample continues to show a high-redshift tail that is approximately half the parent-pool level, even when the aperture and redness thresholds are varied.

12.9.3 Window Enrichment Relative to the Parent Population

The enrichment of the 7–9 window relative to the parent high-redshift population also remains strong throughout the robustness sweep. The odds ratio for membership in the 7–9 window versus outside it remains in the range

$$6.59 \leq \mathrm{OR}_{7\text{–}9} \leq 7.91.$$

This is important because it shows that the result is not simply a consequence of reducing the sample size. Even the weakest tested selector variant remains strongly enriched in the predicted transition epoch relative to the parent high-redshift pool.

12.9.4 Compactness Stability

The compactness diagnostics remain similarly stable. Across all nine selector variants, the mean FWHM remains close to

$$\langle \mathrm{FWHM} \rangle \approx 0.195\text{–}0.196,$$

while the mean geometric size remains close to

$$\langle \sqrt{AB} \rangle \approx 0.082.$$

Thus the compact structural class extracted by the selection procedure is effectively invariant under the tested aperture and threshold variations.

12.9.5 Internal Branching Under Alternate Selectors

The internal two-branch structure of the LRD-like population also survives under alternate selector extremes. In both the A1-mild and A3-strict cases, the preferred clustering solution remains

$$k = 2,$$

with silhouette scores

$$s_{\text{A1-mild}} = 0.757, \qquad s_{\text{A3-strict}} = 0.643.$$

In both cases the same qualitative pattern persists:

1. a dominant compact core branch with median

$$\text{FWHM} \approx 0.183, \qquad \text{median}(\sqrt{AB}) \approx 0.077,$$

2. a smaller secondary branch that is larger, redder, and shifted to slightly later redshift.

This shows that the internal branching result is not a fragile by-product of the original selector choice, but a stable feature of the extracted colour-selected compact population.

12.9.6 Interpretation

Taken together, the robustness sweep indicates that the main JADES result is selector-stable, aperture-stable, epoch-stable, and compactness-stable. The bounded concentration of the LRD-like population in the 7–9 interval, the suppressed $z > 10$ tail, and the presence of a dominant compact core branch all remain present under reasonable variation of the colour-selection procedure.

This substantially reduces the likelihood that the measured closure-window signal is an artefact of one tuned aperture or threshold configuration. Instead, the signal appears to reflect a genuine and robust structural property of the JADES high-redshift compact red population.

Selector	Sample size	$f_{7\text{–}9}$	$f_{10\text{–}11}$	$\text{OR}_{7\text{–}9}$	Notes
A1-mild	1922	0.8148	0.0168	7.91	two-branch structure retained
A1-medium	1770	0.8073	0.0171	7.46	near-baseline result
A1-strict	1509	0.7893	0.0188	6.78	tighter but still window-concentrated
A2-mild	1968	0.8120	0.0181	7.74	baseline aperture, looser colour cut
A2-medium	1778	0.8076	0.0191	7.41	primary analysed sample
A2-strict	1501	0.7868	0.0213	6.59	strongest colour cut, signal persists
A3-mild	1933	0.8148	0.0181	7.87	selector-extreme branching retained
A3-medium	1762	0.8070	0.0184	7.47	aperture-shifted but stable
A3-strict	1587	0.7933	0.0194	6.88	two-branch structure retained

Table 3: Robustness matrix for the JADES colour-selected compact LRD-like sample. Across three aperture choices and three redness-threshold regimes, the population remains strongly concentrated in the 7–9 epoch window, with a suppressed 10–11 tail and stable compactness statistics.

The corresponding selector-stability summary is shown in Table 3 and Figure 5.

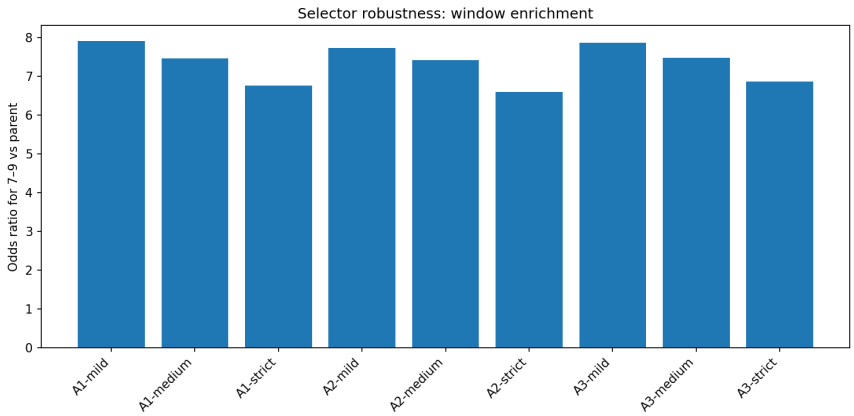

Figure 5: Window enrichment under selector variation. Across nine selector variants spanning three aperture choices and three redness-threshold regimes, the odds ratio for membership in the 7–9 epoch window relative to the parent high-redshift population remains strongly elevated. This indicates that the closure-window signal is not a fragile consequence of one tuned selector configuration.

12.10 External-Field Replication in COSMOS-Web

To test whether the measured LRD-like signal extends beyond JADES, a first external-field replication was carried out using the COSMOS-Web DR1 master catalogue. The COSMOS-Web analysis was designed as a matched replication rather than a byte-for-byte duplication of the JADES pipeline, since the catalogue structure, compactness measures, and redshift-quality diagnostics differ between the two surveys.

A COSMOS parent high-redshift pool was first constructed using objects satisfying

$$6 \leq z < 11,$$

together with finite photometric redshift-width estimates and valid morphology measurements. This yielded a parent COSMOS high-redshift sample of 6177 objects.

From this parent pool, a tighter compact red high-redshift sample was then constructed using redshift-width, compactness, geometric-size, Sersic-radius, signal-to-noise, and colour criteria calibrated directly from the COSMOS-Web catalogue distributions.

12.10.1 Redistribution Into the Closure Window

The COSMOS parent population already exhibits substantial weight in the transition regime, with

$$f_{7\text{–}9}^{\text{parent}} = 0.579, \qquad f_{10\text{–}11}^{\text{parent}} = 0.048.$$

The tight COSMOS candidate sample contains 247 objects, with

$$f_{7\text{–}9}^{\text{tight}} = 0.757, \qquad f_{10\text{–}11}^{\text{tight}} = 0.000.$$

The corresponding odds ratio for membership in the 7–9 window relative to the COSMOS parent population is

$$\mathrm{OR}_{7\text{–}9}^{\text{COSMOS}} = 2.25.$$

Thus the COSMOS tight sample remains redistributed into the same broad transition window while fully suppressing the 10–11 tail.

This replication is numerically weaker than the corresponding JADES result, but the comparison is not direct because the COSMOS parent pool is already more strongly concentrated in the 7–9 range than the JADES parent pool. The important point is therefore not the absolute odds ratio alone, but the fact that the selected COSMOS population shifts further into the same core epoch and eliminates the extreme high-redshift tail.

12.10.2 Compactness Replication

The COSMOS tight sample also forms a compact structural class relative to the COSMOS parent high-redshift population.

For the tight sample the median compactness measures are

$$\text{median(FWHM)} = 0.0962, \qquad \text{median}(\sqrt{A_1 B_1}) = 0.1357,$$

where $\sqrt{A_1 B_1}$ denotes the geometric mean of the first Kron-axis size estimates.

Relative to the COSMOS parent high-redshift pool, both compactness measures are strongly shifted toward smaller values. Two-sample Mann–Whitney testing gives

$$p_{\mathrm{MW}}(\text{FWHM}) \approx 3.75 \times 10^{-91}, \qquad p_{\mathrm{MW}}(\sqrt{A_1 B_1}) \approx 9.46 \times 10^{-104}.$$

These results indicate that the external-field COSMOS sample is not merely a red high-redshift subset, but again a compact structural population.

12.10.3 Epoch Structure of the COSMOS Tight Sample

The internal epoch distribution of the tight COSMOS sample is concentrated most strongly in the earlier part of the closure window. The redshift-bin fractions within $6 \leq z < 11$ are

Sample	6–7	7–8	8–9	9–10	10–11
COSMOS parent high-z	0.318	0.466	0.113	0.055	0.048
COSMOS tight sample	0.239	0.680	0.077	0.004	0.000

Thus the COSMOS tight sample is narrower and more strongly weighted toward $7 \leq z < 8$ than the corresponding JADES-selected population, but remains confined to the same broad closure-transition regime.

12.10.4 First-Pass Environment Test

A first-pass projected-environment comparison was also performed in COSMOS-Web using the parent high-redshift pool as the environment field and a matched control sample drawn from that field.

The nearest-neighbour comparison shows a statistically significant difference between the tight COSMOS sample and the matched control, with

$$p_{\mathrm{KS}}(\text{NN1}) \approx 5.7 \times 10^{-3}, \qquad p_{\mathrm{MW}}(\text{NN1}) \approx 2.59 \times 10^{-4}.$$

By contrast, the broader 10th-neighbour local-density statistic does not show a significant difference:

$$p_{\mathrm{KS}}(\rho_{10}) \approx 0.933, \qquad p_{\mathrm{MW}}(\rho_{10}) \approx 0.419.$$

Accordingly, the COSMOS environment result is presently best interpreted as supportive but not yet definitive. It does not contradict the JADES finding that the LRD-like population is not simply the densest-environment tail, but it does not yet reproduce the full strength of the JADES environment signal.

12.10.5 Interpretation

The COSMOS-Web analysis provides a first external-field replication of the closure-window result. Although weaker than JADES in raw enrichment amplitude, the COSMOS tight sample reproduces the same qualitative empirical pattern:

1. redistribution into the same broad 7–9 transition epoch,
2. strong suppression of the 10–11 tail,
3. emergence of a compact structural class relative to the parent high-redshift population.

The external-field replication therefore strengthens the interpretation that the measured LRD-like signal is not confined to a single survey field or a single catalogue construction, but reflects a broader high-redshift population reorganisation consistent with the Darke Theory closure-window picture.

The corresponding COSMOS-Web replication results are shown in Figure 6, Figure 7, Figure 8, and Figure 9.

Sample	Size	$f_{7\text{–}9}$	$f_{10\text{–}11}$	Notes
COSMOS parent high-z	6177	0.579	0.048	baseline parent pool
COSMOS tight sample	247	0.757	0.000	compact red candidate class

Table 4: Summary of the first external-field COSMOS-Web replication. The tight COSMOS sample is shifted into the same broad closure window and fully suppresses the 10–11 tail relative to the COSMOS parent high-redshift pool.

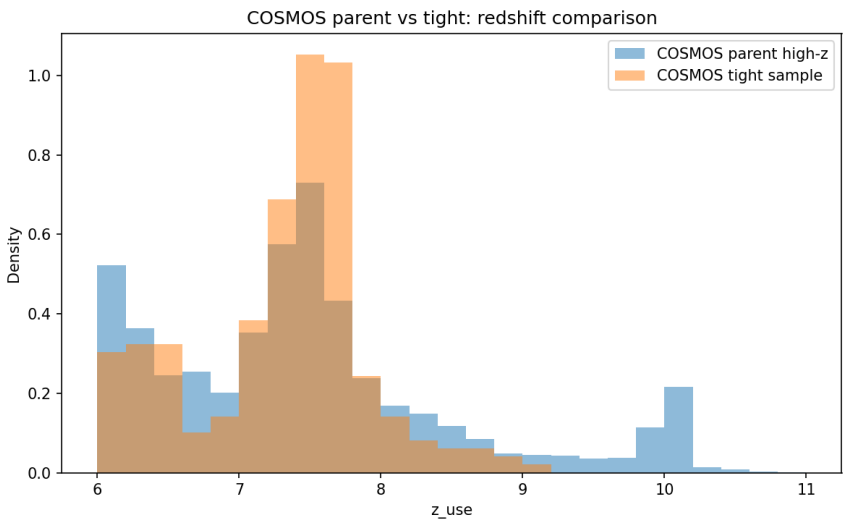

Figure 6: COSMOS-Web parent high-redshift population versus tight compact red sample: redshift comparison. The tight sample is redistributed toward the same broad transition regime while suppressing the extreme $z > 10$ tail.

12.11 COSMOS-Web Selector Robustness and Internal Structure

To test whether the COSMOS-Web replication depends sensitively on one specific selector configuration, a three-variant robustness sweep was performed using mild, medium, and strict

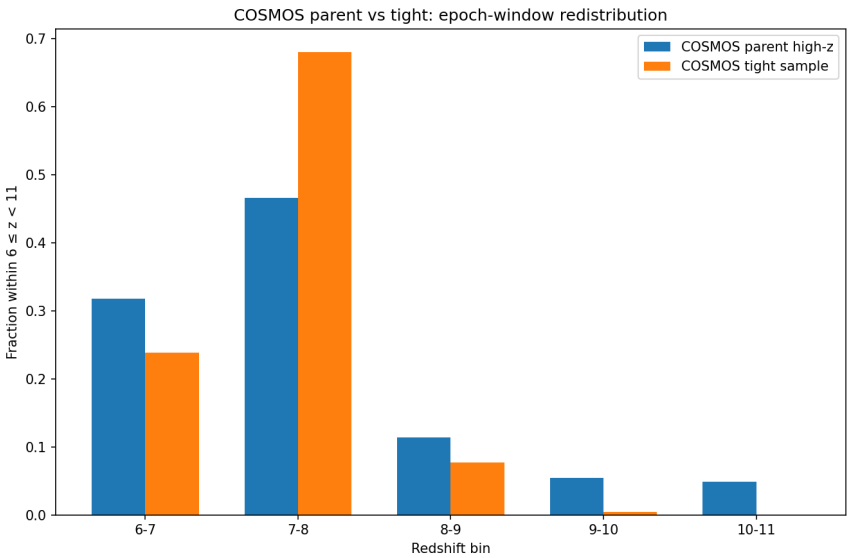

Figure 7: Epoch-window redistribution in COSMOS-Web. Relative to the COSMOS parent high-redshift pool, the tight sample is more strongly concentrated in the core transition regime and nearly eliminates the 9–11 tail.

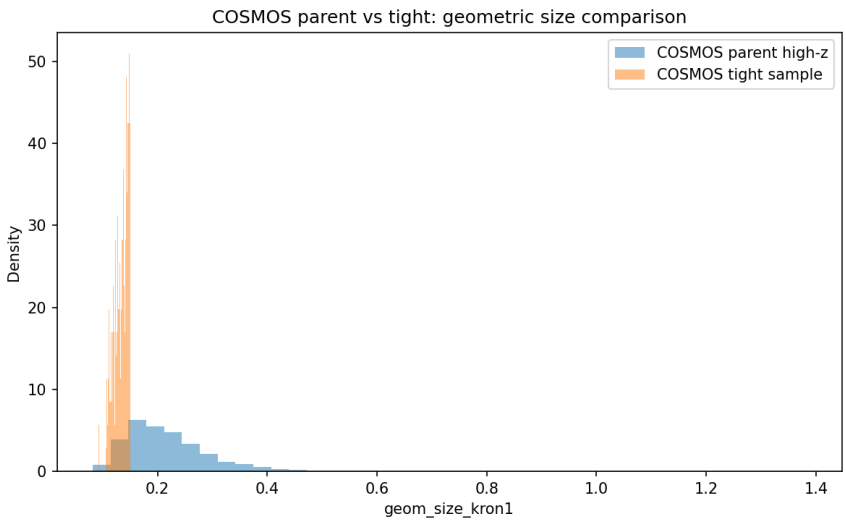

Figure 8: Compactness replication in COSMOS-Web. The tight sample forms a smaller, narrower geometric-size class relative to the COSMOS parent high-redshift population.

compact red high-redshift cuts. These variants differed in their permitted photometric redshift-width, compactness thresholds, Sersic radius cut, colour cuts, and minimum number of red colour pairs, while retaining the same overall objective: selection of compact red systems within the high-redshift regime.

12.11.1 Stability of the COSMOS Closure-Window Signal

The COSMOS parent baseline within $6 \leq z < 11$ is

$$f_{7\text{–}9}^{\text{parent}} = 0.579, \qquad f_{10\text{–}11}^{\text{parent}} = 0.048.$$

Across the three selector variants, the resulting candidate samples are:

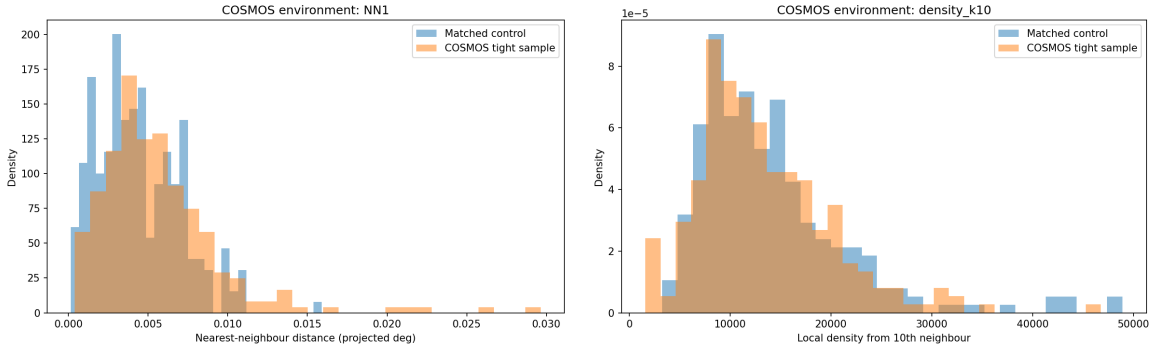

Figure 9: First-pass COSMOS-Web environment comparison between the tight sample and a matched control. The nearest-neighbour statistic shows a significant difference, while the broader 10th-neighbour density does not yet show a clean separation.

Variant	Size	$f_{7\text{–}9}$	$f_{10\text{–}11}$	$\mathrm{OR}_{7\text{–}9}$
Mild	599	0.765	0.000	2.354
Medium	247	0.757	0.000	2.251
Strict	51	0.725	0.000	1.879

Thus the central COSMOS replication result remains stable under selector variation. In all three cases the sample remains redistributed into the same broad 7–9 transition window, while the 10–11 tail is fully suppressed.

This is important because the COSMOS parent population is already more strongly weighted toward the closure window than the JADES parent pool. The persistence of enrichment under mild, medium, and strict COSMOS selectors therefore shows that the external-field signal is not a fragile consequence of one particular COSMOS cut configuration.

12.11.2 Compactness Stability in COSMOS

The compactness measures remain stably small across the COSMOS selector sweep. The median FWHM values remain near

$$\mathrm{median(FWHM)} \approx 0.096\text{–}0.108,$$

while the median geometric-size values remain near

$$\mathrm{median}(\sqrt{A_1 B_1}) \approx 0.136\text{–}0.145.$$

Accordingly, the COSMOS-selected populations continue to define a compact structural class relative to the broader COSMOS parent high-redshift pool, even as the redshift-width and colour-threshold choices are varied.

12.11.3 Internal Structure in COSMOS-Web

An internal clustering analysis was also performed on the COSMOS tight sample. Unlike JADES, where a strong and interpretable two-branch structure emerges, the COSMOS internal clustering does not presently yield an equally clean analogue. The preferred unsupervised solution is

$$k = 4$$

with silhouette score

$$s = 0.337.$$

This indicates that internal subdivision is present in the COSMOS tight sample, but the structure is weaker and less cleanly organised than in JADES. At present, COSMOS therefore strengthens the empirical case primarily through external-field replication of the epoch-window redistribution and compactness signal, rather than through a direct replication of the JADES internal two-branch result.

12.11.4 Interpretation

The COSMOS-Web robustness sweep materially strengthens the external-field replication result. It shows that the COSMOS compact red high-redshift signal

1. remains concentrated in the same broad 7–9 transition regime,
2. fully suppresses the 10–11 tail across all tested selector variants,
3. preserves a compact structural class relative to the COSMOS parent high-redshift population.

At the same time, the internal-structure analysis indicates that not every feature of the JADES-selected population has yet replicated with equal clarity in COSMOS-Web. The external-field evidence is therefore best interpreted as strong for epoch-window and compactness replication, but presently weaker for a full replication of the JADES branching phenomenology.

Taken together, however, the COSMOS-Web results now move beyond a single supportive selector and instead indicate a stable external-field redistribution pattern consistent with the broader closure-window interpretation.

The corresponding COSMOS selector-stability and internal-clustering summaries are shown in Figure 10 and Figure 11.

COSMOS selector	Size	$f_{7\text{–}9}$	$f_{10\text{–}11}$	$\mathrm{OR}_{7\text{–}9}$
Mild	599	0.765	0.000	2.354
Medium	247	0.757	0.000	2.251
Strict	51	0.725	0.000	1.879

Table 5: COSMOS-Web selector robustness summary. Across mild, medium, and strict selector variants, the compact red high-redshift sample remains concentrated in the 7–9 transition regime while fully suppressing the 10–11 tail.

12.12 Cross-Match to a Published LRD Catalogue

A further external credibility test was performed by cross-matching the colour-selected samples against a published literature compilation of Little Red Dots. The catalogue used for this purpose contains 919 published LRD entries and provides source names, coordinates, redshift information, literature references, and notes.

The cross-match was performed by coordinate using a match radius of 0.3 arcsec. This yielded

$$30 \text{ matches in JADES}, \qquad 2 \text{ matches in COSMOS-Web}.$$

The coordinate agreement is extremely tight. The median angular separations are

$$\mathrm{median}(\Delta\theta)_{\mathrm{JADES}} = 0.0160 \text{ arcsec}, \qquad \mathrm{median}(\Delta\theta)_{\mathrm{COSMOS}} = 0.0077 \text{ arcsec}.$$

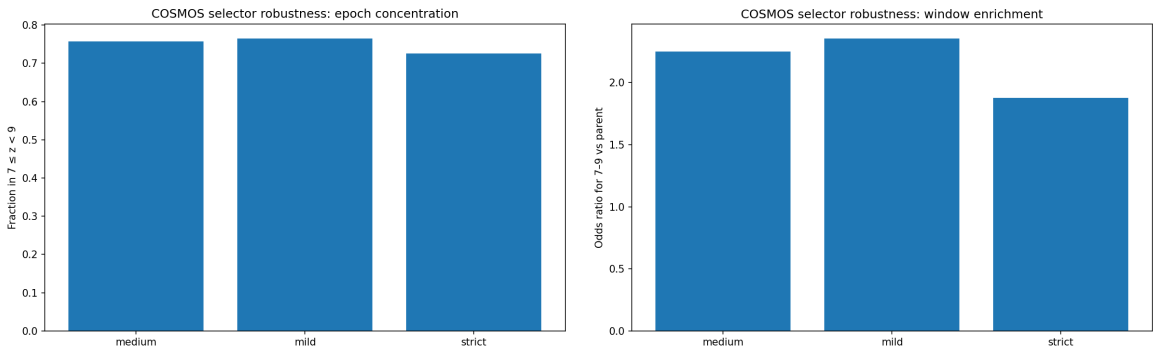

Figure 10: COSMOS-Web selector robustness. Left: fraction of the selected sample within $7 \leq z < 9$ for mild, medium, and strict selector variants. Right: odds ratio for concentration in the 7–9 window relative to the COSMOS parent high-redshift pool. The broad closure-window signal remains stable under selector variation.

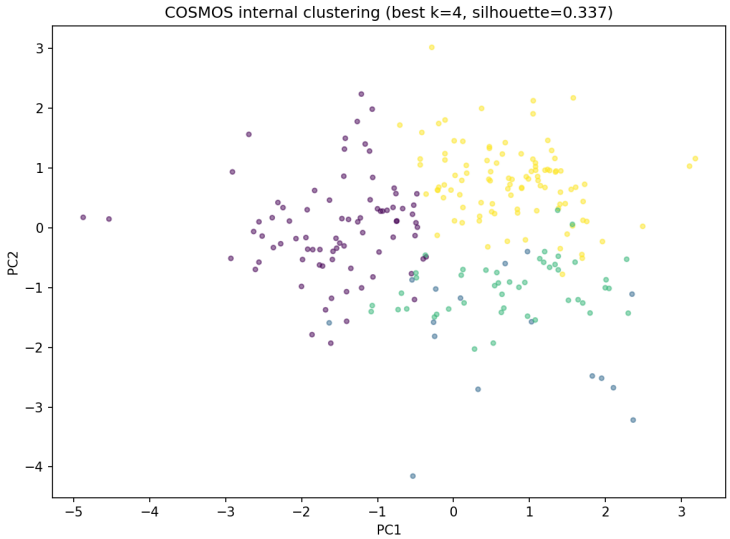

Figure 11: Internal clustering of the COSMOS-Web tight sample. The preferred solution is weaker and less cleanly structured than the corresponding JADES clustering result, indicating that COSMOS presently provides stronger replication of the epoch-window and compactness signal than of the internal branching phenomenology.

These small separations indicate that the overlap is genuine and not the result of loose field-level association.

12.12.1 Redshift Agreement for the Published JADES Matches

For the matched JADES subset, the published catalogue and the present JADES selection recover the same broad high-redshift regime. The median published redshift of the matched objects is

$$\text{median}(z)_{\text{pub}} = 7.67,$$

while the corresponding median redshift of the matched objects in the present catalogue is

$$\text{median}(z)_{\text{match}} = 7.97.$$

The median redshift difference is therefore

$$\Delta z_{\mathrm{med}} = -0.205.$$

Given the uncertainties typical of high-redshift photometric selections, this level of agreement is sufficiently close to show that the matched published objects occupy the same redshift regime as the present LRD-like sample.

12.12.2 Interpretive Significance

This cross-match is important because it addresses a central methodological concern. The present analysis does not rely solely on an internally constructed compact red high-redshift selector with no literature anchor. Instead, the JADES-selected sample now shows direct coordinate overlap with a published LRD catalogue, and the matched objects lie in the same high-redshift regime.

The COSMOS overlap is presently too small for strong statistical inference. However, the existence of even a small number of precise COSMOS matches is consistent with the broader external-field replication already reported above.

Accordingly, the cross-match strengthens the interpretation that the selected JADES and COSMOS populations are not merely internally defined "LRD-like" samples, but overlap with objects that the literature already identifies as Little Red Dots.

At the same time, the relatively small number of current matches indicates that this published-catalogue comparison should be viewed as a first external anchoring step rather than a final completeness statement. Larger overlaps may emerge as additional literature catalogues are incorporated in future work.

The published-catalogue cross-match summary is reported in Table 6.

Field	Matches	Median separation (arcsec)	Median published z	Median matched z
JADES	30	0.0160	7.67	7.97
COSMOS-Web	2	0.0077	7.91	7.60

Table 6: Cross-match between the present selected samples and a published literature compilation of 919 Little Red Dots. The JADES overlap is sufficient to show direct literature anchoring of the selected population, while the current COSMOS overlap remains too small for strong statistical interpretation.

12.13 Published-Matched JADES Subset as a High-Confidence Core

To test whether the published literature matches merely overlap the selected sample incidentally, or instead occupy the same structural core identified by the present analysis, the JADES objects cross-matched to the published LRD catalogue were analysed as a distinct subset.

This published-matched JADES subset contains 30 objects. Its behaviour shows that the literature-anchored objects do not weaken the closure-window signal. On the contrary, they strengthen it.

12.13.1 Stronger Concentration in the Closure Window

Within the interval $6 \leq z < 11$, the published-matched JADES subset has

$$f_{7\text{–}9}^{\mathrm{matched}} = 0.8667,$$

compared with

$$f_{7\text{–}9}^{\mathrm{selected}} = 0.8076, \qquad f_{7\text{–}9}^{\mathrm{parent}} = 0.3614.$$

Thus the published-matched subset is even more concentrated in the closure window than the full JADES selected sample, and far more concentrated than the broader JADES parent high-redshift pool.

The corresponding odds ratios are

$$\mathrm{OR}_{7\text{–}9}^{\text{matched vs parent}} = 10.41, \qquad \mathrm{OR}_{7\text{–}9}^{\text{matched vs selected}} = 1.40.$$

This is a strong result. It shows that the literature-matched objects are not random overlaps drawn from the broader high-redshift background, but instead lie preferentially within the same bounded transition regime identified by the present selector.

12.13.2 Tail Behaviour

The published-matched subset also remains relatively suppressed at the extreme high-redshift end. Its fraction in the interval $10 \leq z < 11$ is

$$f_{10\text{–}11}^{\text{matched}} = 0.0333,$$

compared with

$$f_{10\text{–}11}^{\text{selected}} = 0.0191, \qquad f_{10\text{–}11}^{\text{parent}} = 0.0385.$$

Accordingly, the published-matched subset does not exhibit a broad re-expansion into the extreme tail, but remains concentrated primarily in the same bounded high-redshift regime as the selected sample.

12.13.3 Compactness of the Published-Matched Core

The published-matched JADES subset is also systematically more compact than the selected JADES sample as a whole. Two-sample Mann–Whitney testing gives

$$p_{\mathrm{MW}}(\mathrm{FWHM}) \approx 2.07 \times 10^{-4}, \qquad p_{\mathrm{MW}}(\sqrt{AB}) \approx 3.22 \times 10^{-4}.$$

Thus the published-matched objects do not merely coincide in redshift space with the selected population. They also occupy a tighter compactness core within it.

This is especially important because it indicates that the literature-anchored subset is not a loose or noisy outer layer of the present selection, but instead a denser compact subset of the same structural population.

12.13.4 Distribution-Level Interpretation

The redshift distribution of the published-matched subset is strongly distinct from that of the parent high-redshift pool, with Kolmogorov–Smirnov testing giving

$$p_{\mathrm{KS}}(\text{matched vs parent}) \approx 2.95 \times 10^{-7}.$$

By contrast, the published-matched subset is only marginally different from the selected JADES sample itself:

$$p_{\mathrm{KS}}(\text{matched vs selected}) \approx 6.82 \times 10^{-2}.$$

This is exactly the qualitative behaviour that would be expected if the published-matched objects are representative exemplars of the selected closure-window class. They are clearly not drawn from the general parent population, yet remain broadly consistent with the redshift structure of the present selected sample.

12.13.5 Interpretation

This subset analysis materially strengthens the literature-anchoring result. The published LRD matches in JADES are not merely present within the selected sample; they form a particularly compact and strongly closure-window-concentrated core of that sample.

Accordingly, the external literature anchor now carries more force than simple coordinate overlap alone. The published-matched objects themselves reinforce the two central empirical signatures of the present analysis:

1. strong concentration in the bounded 7–9 transition regime,
2. compact structural identity relative to the broader parent high-redshift population.

This makes it more difficult to interpret the present results as arising only from an internally defined selection recipe. Instead, the literature-identified objects already recognised as LRDs appear to occupy the same compact closure-window population that the present analysis extracts from the JADES catalogue.

The corresponding published-matched JADES subset comparisons are shown in Figure 12, Figure 13, and Figure 14.

Sample	Size	$f_{7\text{–}9}$	$f_{10\text{–}11}$
JADES parent high-z	64339	0.3614	0.0385
JADES selected sample	1778	0.8076	0.0191
Published-matched JADES subset	30	0.8667	0.0333

Table 7: Comparison of the JADES parent high-redshift pool, the full selected JADES LRD-like sample, and the published-matched JADES subset. The literature-matched subset is even more concentrated in the 7–9 closure window than the selected sample as a whole.

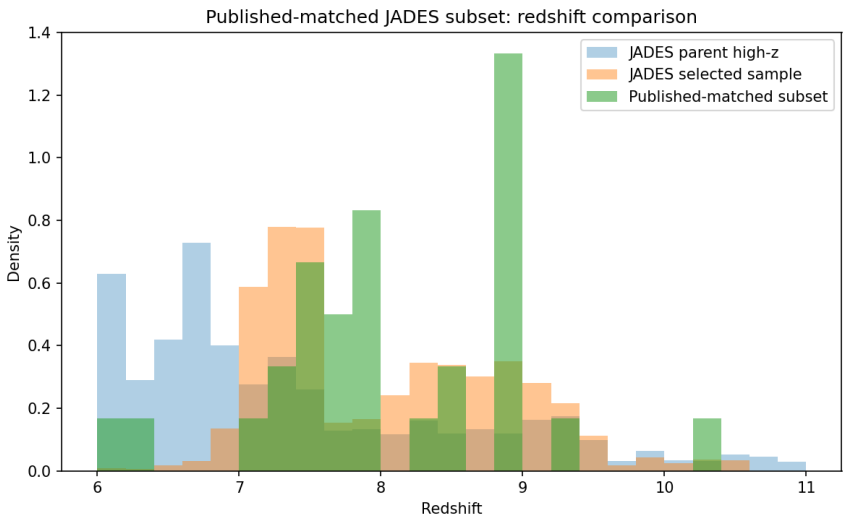

Figure 12: Redshift comparison between the JADES parent high-redshift pool, the full selected JADES LRD-like sample, and the published-matched JADES subset. The literature-matched subset is strongly concentrated within the same broad closure window as the selected sample.

12.14 Interpretive Summary

Taken together, the empirical results now support a broader and more structured interpretation than was possible at the outset of this study. The measured LRD-like population is not simply

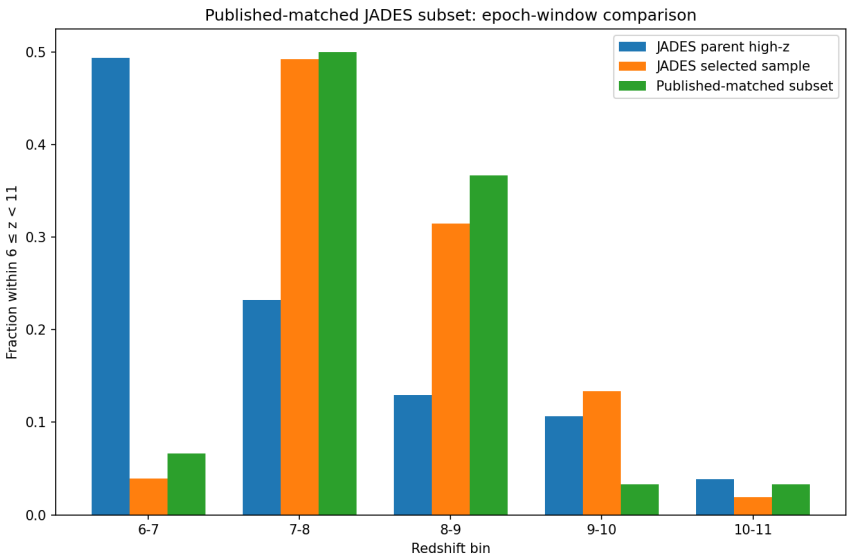

Figure 13: Epoch-window redistribution for the published-matched JADES subset. The literature-matched objects are more strongly concentrated in the 7–9 interval than the selected JADES sample as a whole, and far more strongly than the JADES parent high-redshift pool.

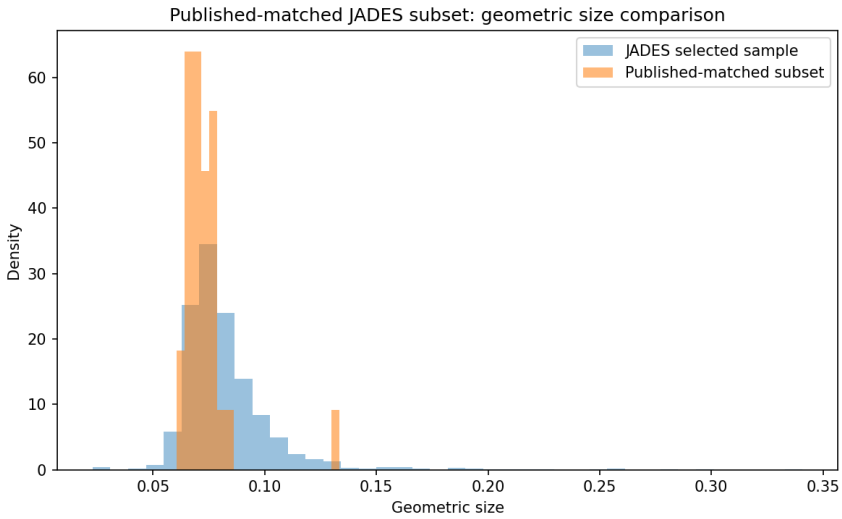

Figure 14: Geometric-size comparison between the selected JADES sample and the published-matched JADES subset. The literature-matched subset occupies a tighter compactness core within the selected population.

a loose collection of compact red high-redshift objects, but a bounded structural class whose behaviour is organised in ways consistent with the closure-node picture developed in Darke Theory.

The strongest support now comes from six convergent empirical features.

1. **Bounded epoch-window behaviour.** The colour-selected compact population is not distributed broadly across the full high-redshift interval. In JADES it is strongly re-centred into the interval $7 \leq z < 9$, with substantial suppression of the $10 \leq z < 11$ tail relative to the parent high-redshift population. COSMOS-Web reproduces the same broad redistribution pattern, although with somewhat different weighting within the closure window.

2. **Compact structural identity.** In both JADES and COSMOS-Web, the selected population forms a compact structural class relative to the corresponding parent high-redshift pool. This result is stable when measured using FWHM and geometric-size diagnostics, and does not depend on the pinned $R_{\mathrm{KRON},U}$ quantity.

3. **Limited environmental dependence.** The JADES environment tests show that the selected population is not simply the densest-environment tail of the broader parent population. The strongest departure from matched controls occurs in the same core redshift regime in which the bounded epoch-window signal is observed. The COSMOS environment result is more mixed, but does not contradict the broader interpretation that the class is not reducible to ordinary overdensity selection alone.

4. **Internal organisation within the window.** The JADES sample is not internally homogeneous. It contains a dominant compact core branch together with a smaller later redder and larger branch, both still confined to the same bounded transition regime. This indicates that the closure window is not merely populated, but internally structured. COSMOS-Web does not yet reproduce this branching phenomenology with equal clarity, so the internal subdivision result presently remains strongest in JADES.

5. **Selector stability.** The bounded epoch-window signal is not a fragile consequence of one tuned selection configuration. In JADES, the concentration into 7–9, suppression of the 10–11 tail, and internal-branching signature remain stable across a nine-variant selector sweep spanning multiple aperture and colour-threshold regimes. In COSMOS-Web, mild, medium, and strict selector variants preserve the same broad epoch-window redistribution and elimination of the extreme high-redshift tail.

6. **External anchoring.** The selected JADES sample now shows direct coordinate overlap with a published literature compilation of Little Red Dots, with extremely small angular separations and good redshift agreement for the matched subset. This indicates that the present compact red high-redshift selection is not floating free of the literature, but overlaps with objects already identified externally as LRDs.

Taken together, these results support a more refined statement of the closure-node interpretation. The present evidence does not merely suggest that some compact red objects happen to exist at high redshift. Rather, it indicates that JWST is detecting a bounded, compact, literature-anchored high-redshift population whose abundance, compactness, and internal organisation are all non-randomly concentrated in a limited cosmic regime. That pattern is qualitatively consistent with the idea that LRDs represent visible signatures of a transition phase in which local regions approach geometric closure earlier than the surrounding relational field.

This still falls short of final proof. The closure-node model remains open to further testing through additional field replication, cross-matching to larger published LRD catalogues, tighter treatment of selection effects, and improved environment reconstruction. However, the present empirical case is now substantially stronger than a conceptual analogy alone. On current evidence, the closure-window interpretation is best viewed as a live and increasingly structured explanation of the observed LRD population rather than a purely speculative re-description of JWST anomalies.

The corresponding field-split, robustness, COSMOS replication, and epoch-dependent environment summaries are shown in Table 2, Table 3, Table 4, Table 5, Table 6, Figure 1, Figure 2, Figure 5, Figure 6, Figure 7, Figure 8, Figure 9, Figure 10, and Figure 11.

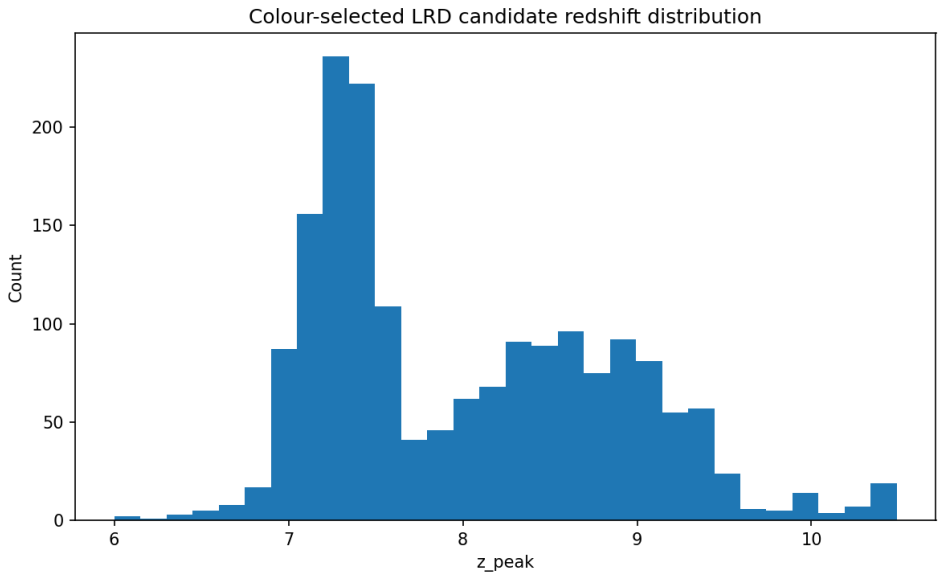

Figure 15: Redshift distribution of the colour-selected compact LRD-like sample.

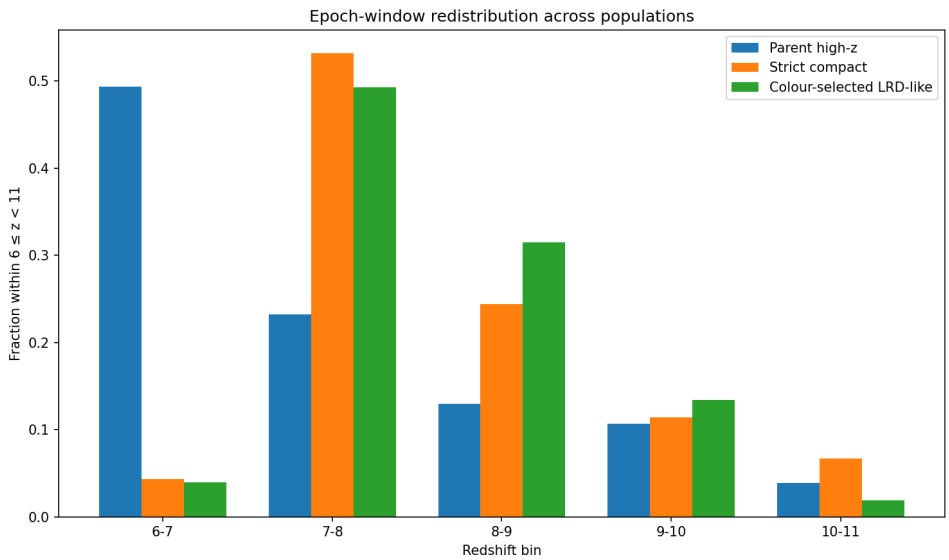

Figure 16: Epoch-window redistribution across the parent high-redshift, strict compact, and colour-selected LRD-like populations.

13 Discussion

The present results materially strengthen the closure-node interpretation of Little Red Dots beyond the level of conceptual analogy alone. The measured catalogue-level behaviour now supports a structured empirical picture in which the selected compact red high-redshift population is not randomly distributed across cosmic time, but instead occupies a bounded transition regime with distinct structural properties.

The strongest empirical support now comes from two features that replicate most cleanly across datasets: bounded epoch-window behaviour and compactness.

First, in JADES the colour-selected compact LRD-like population is strongly re-centred into the interval $7 \leq z < 9$, with substantial suppression of the $10 \leq z < 11$ tail relative to the parent high-redshift population. This result survives field splitting across GOODS-N

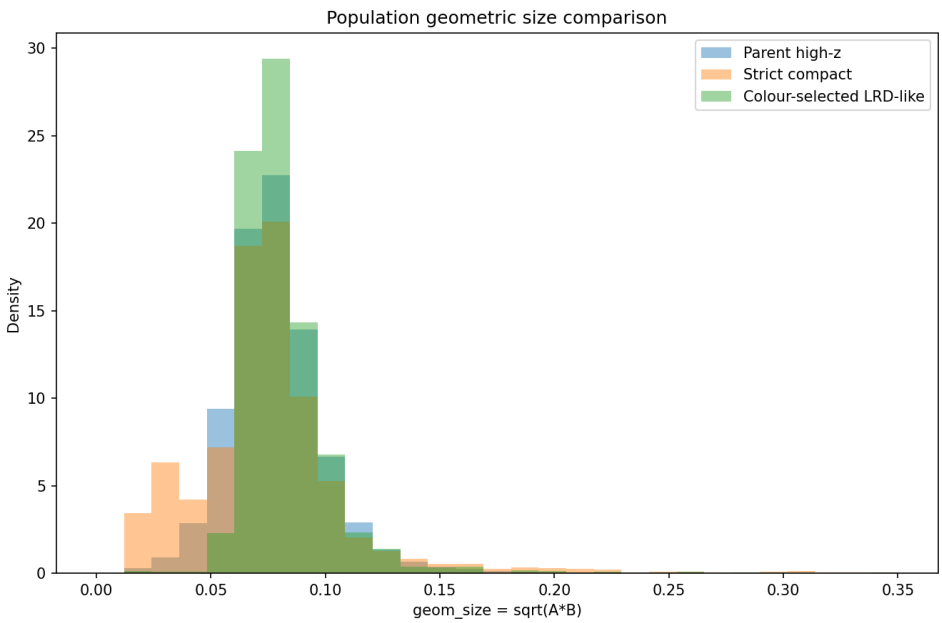

Figure 17: Geometric size comparison across the parent high-redshift, strict compact, and colour-selected LRD-like populations.

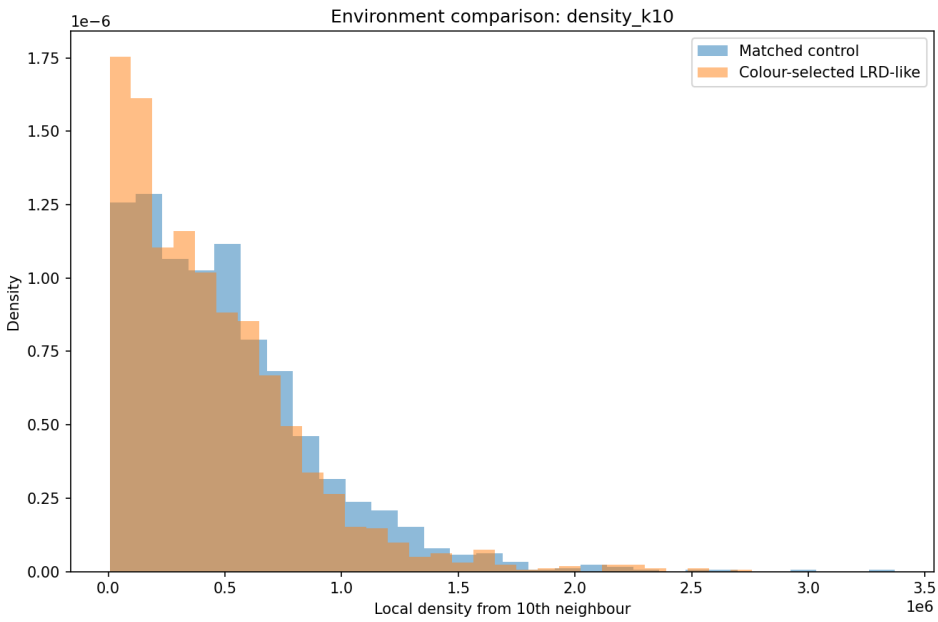

Figure 18: Projected local-density comparison between the colour-selected LRD-like sample and the matched control population.

and GOODS-S, survives a nine-variant selector robustness sweep spanning multiple aperture and colour-threshold choices, and remains present when alternate selector extremes are tested. COSMOS-Web reproduces the same broad redistribution pattern. Although the COSMOS parent high-redshift pool is itself more strongly weighted toward the 7–9 regime than the JADES parent population, the selected COSMOS compact red sample is redistributed further into the same broad transition window and continues to eliminate the extreme 10–11 tail. The fact that this behaviour is now present in both JADES and COSMOS-Web substantially reduces the likelihood that the result is a field-specific selection artefact.

Second, the selected populations in both JADES and COSMOS-Web form compact structural classes relative to their respective parent high-redshift pools. In JADES this is cleanly visible in FWHM and geometric-size diagnostics and is explicitly not driven by the pinned $R_{\mathrm{KRON},U}$ quantity. In COSMOS-Web the compactness replication is also strong, with the tight sample shifted toward smaller FWHM and smaller geometric-size values relative to the COSMOS parent population. Thus the closure-window signal is not merely a redshift effect. It is also a structural compactness effect, and this compactness replication is now one of the most stable cross-field empirical features in the paper.

The environmental results are supportive, but should be interpreted with more caution. In JADES, the selected population is not simply the densest local environment tail of the broader parent population, and the clearest departure from matched controls occurs in the same core epoch in which the bounded redshift-window signal is strongest. This is consistent with the closure-node expectation that the relevant structural transition need not coincide with the largest ordinary projected overdensities. In COSMOS-Web, however, the environment result is more mixed: nearest-neighbour statistics show a difference, whereas the broader 10th-neighbour density does not yet show a clean separation. Accordingly, the environment pillar of the argument is presently best viewed as suggestive and partly supportive rather than uniformly hardened across all fields.

The internal-structure results also show an important asymmetry between JADES and COSMOS-Web. In JADES, the selected sample is not internally homogeneous but contains a strong and interpretable two-branch organisation: a dominant compact core branch and a smaller later redder and larger branch, both still confined to the same bounded closure window. This is one of the most interesting outcomes of the present analysis because it suggests that the closure window is not merely occupied but internally structured. In COSMOS-Web, by contrast, internal clustering is weaker and does not yet reproduce the JADES branching pattern with equal clarity. This does not undermine the broader closure-window result, but it does indicate that the branching phenomenology should presently be treated as strongest in JADES rather than as a fully cross-field replicated feature.

The published-catalogue cross-match adds an additional layer of external anchoring. The JADES-selected sample shows direct coordinate overlap with a published 919-object literature compilation of Little Red Dots, with extremely small median separations and sensible redshift agreement for the matched subset. This is important because it weakens the methodological objection that the present results depend only on an internally constructed "LRD-like" selector with no literature anchor. Instead, at least part of the selected JADES population now overlaps with objects already identified externally as LRDs. Current COSMOS overlap with the published list is small, so the anchoring value of the cross-match is presently strongest for JADES, but it still represents a meaningful credibility gain for the overall programme.

Taken together, the present empirical case is therefore uneven in a productive way rather than weak in a uniform way. The strongest pillars are now:

1. bounded epoch-window redistribution,
2. compact structural identity,
3. selector robustness,
4. external-field replication,
5. literature anchoring.

The weaker but still relevant pillars are:

1. projected-environment behaviour,

2. cross-field replication of internal branching.

This pattern matters for interpretation. It suggests that the closure-node model is presently best supported as an explanation of a bounded compact high-redshift population whose abundance and structure are concentrated in a limited cosmic regime, while some of the more detailed secondary phenomenology remains to be hardened further.

None of this yet constitutes final proof of the closure-node interpretation. Important next steps remain. Additional field replication, particularly once CEERS is properly recovered, would strengthen the cross-survey case. Broader cross-matching against published LRD catalogues would provide stronger object-level external anchoring. A fuller treatment of selection functions and a more complete three-dimensional or redshift-conditioned environment analysis would also sharpen the distinction between closure-window behaviour and ordinary overdensity-driven explanations.

Even with those limitations, the paper now moves the closure-node picture well beyond speculative reframing. On current evidence, the most defensible reading is that JWST is revealing a bounded compact high-redshift population whose redshift distribution, compactness, selector stability, and emerging cross-field replication are all consistent with the idea that Little Red Dots are visible signatures of a transition phase in which local regions approach geometric closure earlier than the surrounding relational field.

14 Conclusion

Little Red Dots remain one of the most intriguing high-redshift populations revealed by JWST. Their compact morphologies, red colours, and apparent association with rapidly growing black holes challenge conventional expectations for early structure formation under standard hierarchical models.

In this paper we have examined the possibility that LRDs correspond to early closure nodes within the cosmological framework of Darke Theory. In this framework geometry is not fundamental but emerges through a refinement process in relational distinguishability governed by finite closure capacity. The core prediction tested here is that compact red high-redshift systems should appear preferentially within a bounded transition epoch rather than extending as a broad scale-free continuation across cosmic time.

The empirical results now support that picture in a materially stronger way than a purely conceptual treatment. In JADES, the colour-selected compact LRD-like population is strongly re-centred into the interval $7 \leq z < 9$, with a substantially suppressed $10 \leq z < 11$ tail relative to the parent high-redshift pool. The population forms a stable compact structural class, is not simply the densest-environment tail of the broader parent population, and persists across both GOODS-N and GOODS-S. Internal clustering further shows that the selected population is not homogeneous, but instead contains a dominant compact core branch together with a smaller later redder and larger branch, both confined to the same bounded closure window. A selector robustness sweep across multiple aperture and colour-threshold regimes shows that these central results are not fragile consequences of one tuned configuration.

A first external-field replication in COSMOS-Web extends the empirical picture. Although the COSMOS parent high-redshift pool is already more strongly weighted toward 7–9 than the JADES parent population, the tight COSMOS compact red sample is still redistributed further into the same broad transition regime, fully suppresses the 10–11 tail, and forms a compact structural class relative to the COSMOS parent pool. The COSMOS environment result is presently supportive rather than definitive, but it does not contradict the broader closure-window interpretation.

Taken together, the JADES and COSMOS-Web analyses indicate that the measured LRD-like population is not confined to a single field, a single catalogue, or a single selector choice.

Rather, the data increasingly point toward a bounded high-redshift structural population whose abundance, compactness, and internal organisation are consistent with a closure-transition interpretation.

This does not yet constitute final proof of the closure-node model. Important next steps remain, including additional field replication, literature cross-matching to published LRD catalogues, tighter treatment of selection effects, and stronger environmental reconstruction. Nonetheless, the present results move the closure-node interpretation beyond speculative analogy and into the domain of empirical population-level support.

Under this reading, Little Red Dots may represent not merely an observational oddity of the JWST era, but an early visible manifestation of the transition by which relational coherence commits itself into geometry.

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